

# Intelligent Decision Support System for Sri Lankan Tea Estate Management

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**Abstract** – Sri Lankan tea sector is significant for the country's economy and still suffers from the problems that relate to labor management issues, tea leaves disease, climate variations, and tea quality analysis. All these factors negatively affect the efficiency of tea estate management, productivity, and sustainability of the process. The research provides a concept of the Intelligent Decision Support System for Sri Lankan Tea Estate Management, which includes the application of artificial intelligence, machine learning, deep learning, Internet of Things technologies, and data analytics in a single decision support system. The research presents the four intelligent modules Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea Specific Weather and Climate Intelligence System, and AI Based Tea Sorting and Grading Automation. Labor Management and Welfare Optimization module uses machine learning algorithms to optimize the labor management process and implements IoT wearable devices to assess the health state of workers and their welfare. Tea Leaf Disease Detection and Pattern Analysis module uses the YOLOv8s algorithm to detect and identify tea leaf diseases based on the image dataset. Weather Intelligence module analyzes historical and real time meteorological data to provide personalized weather forecast and climate related recommendations for the estate. Tea Sorting and Grading module provides automated tea quality classification using deep learning image analysis.

**Keywords**— *Artificial Intelligence, Decision Support System, Machine Learning, Deep Learning, Internet of Things, YOLOv8s.*

## I. INTRODUCTION

The tea industry is one of the most valuable agricultural industries in Sri Lanka. It is crucial in terms of the country's economy due to its contribution to exports, job creation and improvement of rural people's life. The Sri Lankan Ceylon tea is known throughout the world for its high quality. However, the modern management of tea estates became quite difficult because of labor shortage, tea leaves disease, changed weather condition and necessity to keep tea's consistent quality. All these factors negatively influence the tea estates' productivity, increase the cost of production and reduce the performance of tea estates.

Most tea estates still rely on manual data keeping and traditional management techniques. The majority of decisions is made according to personal experience and observations that takes much time. Therefore, estate manager may not get necessary information in time. Consequently, there is a need to create some intelligent system that could help to make fast and reasonable decisions.

Recent advances in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Internet of Things (IoT) allow to introduce changes in tea estate management and make it more efficient. With their help, it is possible to analyze great volumes of data, perform various routine processes, control workers' health, detect and analyze tea leaf diseases, forecast weather conditions and increase the accuracy of quality assessment. Combining all these technologies, estate managers will be able to make more informed decisions and thus increase the performance of estate operations.

The purpose of this research is to design an intelligent decision support system for the Sri Lankan tea estate management. The proposed system consists of four intelligent modules, each of which assists in solving certain problems related to tea estate management. Labor Management and Welfare Optimization module uses machine learning algorithms for labor allocation optimization and IoT wearables for continuous health monitoring of the workers. Tea Leaf Disease Detection and Pattern Analysis module uses the YOLOv8s deep learning model to detect and identify tea leaf diseases in tea leaves images. Tea Specific Weather and Climate Intelligence System analyses weather information in order to give localized forecasts and recommendations regarding tea planting. AI Based Tea Sorting and Grading Automation module uses deep learning methods for tea grade classification.

## II. LITERATURE REVIEW

Tea estate management has been examined by researchers to boost productivity and cut management problems. Most studies use Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL) and Internet of Things (IoT) to solve issues in tea cultivation. These studies give results but most of them look at only one problem instead of offering a full tea estate management solution [1].

Many researchers focus on improving labor management in agriculture. Machine learning predicts labor needs, assigns tasks and raises workforce productivity [2],[4],[5]. Studies in Sri Lankan tea estates show that labor shortages, uneven workloads and worker welfare are challenges that affect daily estate operations [7],[8]. The Tea Research Institute of Sri Lanka has also published guidelines to improve labor management and worker welfare in tea estates [10]. Although these studies improve workforce management they do not monitor worker health during activities. The proposed system addresses this limitation by combining machine learning for labor allocation with wearable devices for real-time monitoring of worker heart rate and SpO<sub>2</sub> levels. This approach supports worker welfare while improving estate productivity.

Tea leaf diseases are another challenge because they lower both tea yield and tea quality. Researchers develop deep learning models to detect plant diseases from leaf images with high accuracy [11]. Other studies introduce tea leaf disease detection systems that improve disease identification under field conditions [12]. Deep learning also applies to agricultural image analysis and tea quality identification [13]. However, most existing systems only identify diseases. Do not support other tea estate management activities. The proposed system uses the YOLOv8s model to detect tea leaf diseases and integrates the detection results with treatment recommendations and the overall decision support platform. This enables estate managers to identify diseases and respond more effectively.

Weather conditions have an impact on tea cultivation. Temperature, rainfall, humidity and wind influence plant growth disease spread, irrigation needs and harvest schedules. Previous studies apply machine learning models to predict tea yield and analyze climate changes in tea growing regions [15],[20]. Although these studies give weather predictions most of them focus only on weather forecasting or crop yield estimation. They do not give recommendations that directly help tea estate management. The proposed Tea-Specific Weather and Climate Intelligence module combines weather records with real-time weather data to help estate managers. A Gradient Boosting Regressor predicts weather conditions while a Multi-output Random Forest Regressor estimates tea leaf disease risks based on the predicted weather. The system also gives recommendations for irrigation, fertilizer use, harvesting and disease prevention.

Tea sorting and grading are important because they decide the quality and market value of tea products. Traditional grading relies on inspection, which needs experienced workers and may produce inconsistent results. Recent studies use deep learning to automate tea quality classification and improve grading accuracy [13],[21]. Image-based grading systems also cut processing time while improving grading consistency [22] [23]. Review studies show that deep learning has become an effective technology for tea quality monitoring and grading

[24]. However, these systems mainly focus on grading. Are not linked with labor management, disease detection or weather information. The proposed AI-Based Tea Sorting and Grading Automation module uses the ConvNeXtV2-Tiny deep learning model to classify tea grades automatically and links the grading results with the complete tea estate management system.

Most previous studies solve one problem, such as labor management, disease detection, weather prediction or tea grading. Few studies combine these important functions into one intelligent platform. To fix this gap this research proposes a Decision Support System for Sri Lankan tea estate management. The proposed system combines machine learning for workforce allocation, wearable devices for real-time worker health monitoring the YOLOv8s model for tea leaf disease detection a Gradient Boosting Regressor for weather prediction a Multi-output Random Forest Regressor for disease risk prediction and the ConvNeXtV2-Tiny deep learning model for tea sorting and grading. By integrating these four modules into a single platform the proposed system offers decision support, for estate managers boosts productivity supports worker welfare keeps tea quality high and promotes sustainable tea estate management.

## III. METHODOLOGY

### A. System Overview

The proposed Intelligent Decision Support System for the Management of Sri Lanka's Tea Estates is a Web Based Platform that aims at enhancing the management of tea estates. The system comprises of four major modules which include Labor Management and Welfare Optimization, Detection and Analysis of Tea Leaf Disease Patterns, Tea specific Weather and Climate Intelligence and AI based Tea Sorting and Grading Automation. The data collected from various sources is processed through the system and stored in a database. The system undertakes different functions using the various modules and presents important information on a dashboard.

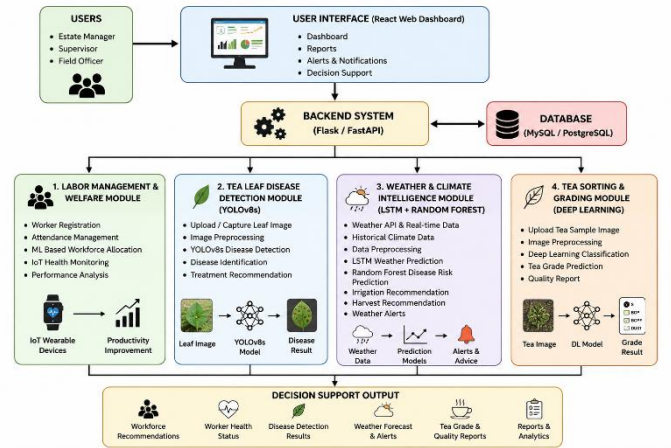


Fig. 1. Overall System Architecture

### B. System Architecture

The design of the system employs a client server architecture that is web based. Managers of estates use the system through a web application created using React.js. In the frontend, one is able to control workers, upload images of tea leaf and tea sample, check weather information and get reports from the system. Requests made by the user are passed on to the backend made using Node.js. The backend interacts with the

database, external APIs and intelligent modules. All data within the system which includes data about workers, the images uploaded, weather information, predictions and reports are saved in the database.

The system has four main modules namely Labor Management and Welfare Optimization, Tea Leaf Disease Detection and Pattern Analysis, Tea-Specific Weather and Climate Intelligence and AI Based Tea Sorting and Grading Automation. Every module handles data individually and passes the results to the dashboard.

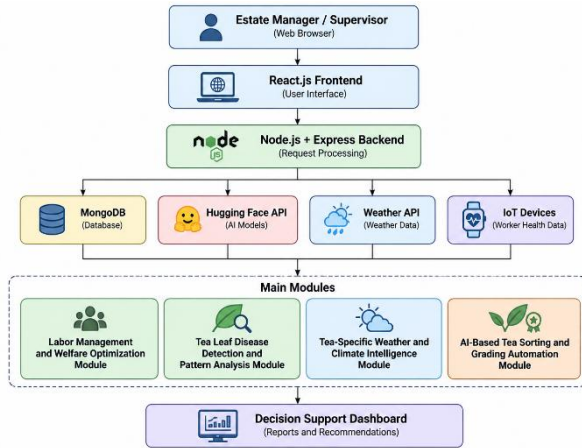


Fig. 2. System Architecture

### C. Labor Management and Welfare Optimization Module

The Workforce Management and Worker Welfare Optimization Module was created to make managing workers better and to help keep workers in condition in the tea fields of Sri Lanka. The module helps managers give the jobs to workers and also watch over their health while they are working.

At the start the manager has to enter all the details about each worker and the list of daily jobs. Details like worker experience, skills, past work and when they are available get stored in the MongoDB database. To suggest the workers for each job a machine learning program looks at the worker details. This way the job assignments get better. The work gets spread more evenly among the workers.

Also the module gets health data from workers in time. Specifically it gets information about the workers heart rate and the amount of oxygen in their blood (SpO<sub>2</sub>) which is tracked by devices while they do their work. This data goes to the Node.js server gets checked and is stored in the MongoDB database. If the heart rate or SpO<sub>2</sub> goes beyond levels the module sends an alert to let people know there might be a problem with a worker.

The front part of the module was built using the React.js program. Makes it easy to manage worker details watch health conditions and see job assignment suggestions. The back part was made using the Node.js and Express.js programs to handle data and connect with devices and the MongoDB database. The results of job assignments and details, about workers' health can be seen on the screen.

### D. Tea Leaf Disease Detection and Pattern Analysis Module

The Tea Leaf Disease Detection and Pattern Analysis Module is used to find out if a tea leaf is in the image that is

uploaded and to spot the diseases that affect the tea leaf. This module helps estate managers find diseases at a time and gives advice on how to treat them to stop the plant from getting hurt.

First the user will upload the tea leaf image using the web application. This image is sent from the React.js frontend to the Node.js backend for processing. After processing the image is sent to the Hugging Face API that uses the YOLOv8s model for image detection.

In YOLOv8s the image will be checked to see if the uploaded image has a tea leaf. If the tea leaf is found the YOLOv8s finds the disease, on the tea leaf. Places it in the correct disease category. Then based on this finding the system gives treatment advice. The results of the disease detection the confidence score and the treatment advice will be sent to the backend. Saved in the MongoDB database. At the end the results are shown on the dashboard.

### E. Tea-Specific Weather and Climate Intelligence Module

The Tea-Specific Weather and Climate Intelligence Module is built to predict weather conditions and tea disease risk for growing regions. The Tea-Specific Weather and Climate Intelligence Module helps tea estate managers plan ahead by warning them of disease risk before it becomes a problem instead of only reacting after damage is visible.

The Tea-Specific Weather and Climate Intelligence Module works by using weather readings of temperature, humidity and rainfall collected for three tea-growing regions: Nuwara Eliya, Kandy and Ratnapura. This data is processed by a Python-based service built with Flask. Two separate models are trained for each region: a Random Forest model that predicts disease risk and a Linear Regression model that predicts temperature. The Tea-Specific Weather and Climate Intelligence Module chose Random Forest for disease risk because tea diseases such as blister blight and grey blight depend on threshold combinations of humidity and temperature. For example risk rises sharply once humidity passes a point and temperature falls in a certain range at the same time. The Tea-Specific Weather and Climate Intelligence Module chose Linear Regression for temperature prediction because temperature changes gradually and smoothly from one hour to the next which a simpler model can capture accurately.

When a request is made the Flask service loads the recent weather reading for the selected region feeds it into the trained models and generates an eight-hour forecast showing predicted temperature and disease risk for each upcoming hour. The Tea-Specific Weather and Climate Intelligence Module returns these predictions along with model accuracy information to the application backend. The backend can then display the information to tea estate managers giving them a region-specific warning of rising disease risk so they can act before an outbreak spreads. Together the historical weather dataset, the trained models and the Flask prediction service form a system that turns raw weather data into forward-looking guidance, for tea disease prevention.

### F. AI-Based Tea Sorting and Grading Automation Module

The AI Based Tea Sorting and Grading Automation Module is made to automate the grading of processed tea by deciding the quality grade. This module can lower the work

needed raise the accuracy of grading and help keep quality assessment the same in tea factories. First the user uploads an image of the processed tea through the web application. The image goes to the backend, which is built with Node.js, from the React.js frontend. The backend sends the image to the Hugging Face API, where a deep learning model looks at the tea sample and puts it into the tea grade.

When the grading is done the prediction and confidence score come back to the backend. Are stored in the MongoDB database. Then the user sees the image the predicted tea grade and the confidence score, on the dashboard. The frontend of this module was built with React.js and the backend uses Node.js and Express.js. Together the web application, database and deep learning model form an efficient way to solve the tea quality grading problem.

#### IV. RESULTS AND DISCUSSION

The proposed Intelligent Decision Support System for Sri Lankan Tea Estate Management was successfully developed and tested as a web based platform. Each module was evaluated separately using its own dataset and trained model. The performance of the system was measured based on the prediction results and the practical use of each module. The results show that the proposed system can support tea estate managers in making better decisions related to labor management, worker health monitoring, tea leaf disease detection, weather based planning, and tea quality grading. This section presents the results of each module and discusses how the system can improve the overall management of tea estates.

##### A. Labor Management and Welfare Optimization Module

The Labor Management and Welfare Optimization Module was developed to monitor worker health and support better workforce management in tea estates. The module uses physiological data collected from IoT wearable devices to identify the stress level of workers. Heart Rate (HR) and Blood Oxygen Saturation (SpO<sub>2</sub>) were selected as the input features because they provide useful information about the physical condition of workers during field activities.

The PhysioNet Non EEG Dataset for Assessment of Neurological Status was used to train the prediction model. The LightGBM Classifier (LGBMClassifier) was trained using Google Colab to classify worker conditions into four categories: Relaxed, Emotional Stress, Cognitive Stress, and Physical Stress. After training, the model achieved an accuracy of 82%, showing good performance in classifying worker stress levels.

The trained model was integrated into the proposed web application. Real time HR and SpO<sub>2</sub> values collected from the IoT device are sent to the prediction model, and the predicted stress level is displayed on the dashboard. This allows estate managers to monitor the health condition of workers during daily activities and quickly identify workers who may require attention.

The experimental results show that the module can successfully combine IoT health monitoring with machine learning to support labor management. By providing real time health information and stress predictions, the system helps estate managers make better decisions while improving worker safety and welfare.

TABLE I  
PERFORMANCE OF THE LABOR MANAGEMENT MODEL

| Parameter      | Value                                                             |
|----------------|-------------------------------------------------------------------|
| Dataset        | PhysioNet (Non-EEG Dataset for Assessment of Neurological Status) |
| Model          | LightGBM Classifier (LGBMClassifier)                              |
| Input Features | Heart Rate (HR), SpO <sub>2</sub>                                 |
| Output Classes | Relaxed, Emotional Stress, Cognitive Stress, Physical Stress      |
| Accuracy       | 82%                                                               |

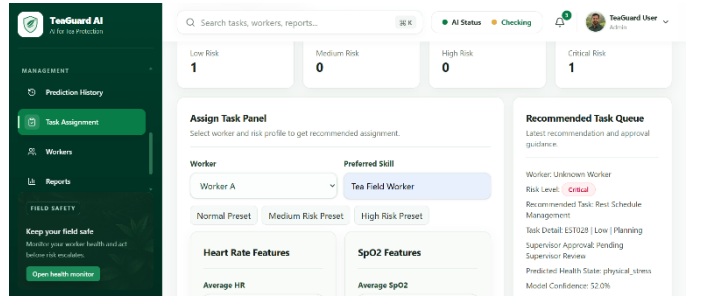


Fig. 3. Labor Management

##### B. Tea Leaf Disease Detection and Pattern Analysis

The Disease Detection and Pattern Analysis module was implemented to detect diseases in tea leaves based on images uploaded to the system. The system first checks the image to determine whether it contains a tea leaf before the disease detection process begins. If it detects a tea leaf, then it passes the image to the disease detection model for the next stage.

This disease detection model was trained using YOLOv8s on a tea leaf disease image dataset. This model can detect various types of tea leaf diseases and display the disease name together with its confidence score. At the end of the prediction process, the system also gives an appropriate recommendation for how to treat the detected disease.

Results from validation tests show that the model obtained 96.1% mAP@0.5 with 95.2% precision, 93.8% recall, and F1-score of 94.5%. This result shows that the disease detection model is capable of accurately detecting diseases in tea leaves in field images. The implementation of the model to the web application enables fast detection of diseases and their early treatment.

TABLE II  
PERFORMANCE OF THE TEA LEAF DISEASE DETECTION MODEL

| Parameter | Value                                                    |
|-----------|----------------------------------------------------------|
| Model     | YOLOv8s                                                  |
| Input     | Tea Leaf Image                                           |
| Output    | Disease Name, Confidence Score, Treatment Recommendation |
| Precision | 95.2%                                                    |
| Recall    | 93.8%                                                    |
| F1-Score  | 94.5%                                                    |

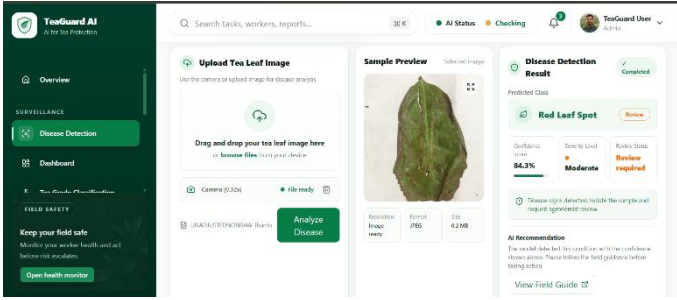


Fig. 4. Tea Leaf Disease Detection

### C. Tea-Specific Weather and Climate Intelligence

The Tea-Specific Weather and Climate Intelligence Module was created to give weather forecasts and climate-based recommendations for people who run tea estates. It pulls weather records and fresh data from the OpenWeatherMap API. To guess temperature the Tea-Specific Weather and Climate Intelligence Module uses a Gradient Boosting Regressor. To estimate the risk of tea leaf diseases under the predicted weather the Tea-Specific Weather and Climate Intelligence Module uses a Multi-output Random Forest Regressor.

The experimental results showed that the Tea-Specific Weather and Climate Intelligence Module reached a prediction accuracy of 89 %. The weather forecasts and disease risk predictions from the Tea-Specific Weather and Climate Intelligence Module were put into a web application that gives recommendations, for irrigation, fertilizer use, harvesting and disease prevention. These recommendations help estate managers make decisions based on changing weather conditions and lower climate-related risks in tea cultivation.

TABLE III

PERFORMANCE OF THE TEA-SPECIFIC WEATHER MODULE

| Parameter                     | Value                                                  |
|-------------------------------|--------------------------------------------------------|
| Weather Prediction Model      | Gradient Boosting Regressor                            |
| Disease Risk Prediction Model | Multi-output Random Forest Regressor                   |
| Weather Data Source           | OpenWeatherMap API                                     |
| Input Features                | Temperature, Humidity, Rainfall, Wind Speed            |
| Output                        | Weather Forecast, Disease Risk, Estate Recommendations |
| Overall Accuracy              | 89%                                                    |

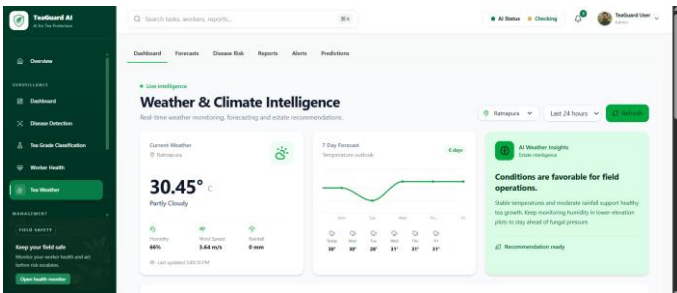


Fig. 5 Tea-Specific Weather and Climate Intelligence

### D. AI-Based Tea Sorting and Grading Automation

The Tea Sorting and Grading AI Model was created for the purpose of classifying the tea into different tea grade categories. The model utilizes the ConvNeXtV2-Tiny (convnextv2\_tiny.fcmae\_ft\_in22k\_in1k) model, which was trained using the image dataset of processed tea. Once an image is uploaded, the model detects the tea grade and shows the result via the web application.

Experimental results show that the model had 99.7% average precision, 99.7% average recall, and 99.7% average F1-score. In general, the majority of the tea grade categories were detected properly, which means that the model has a great capability to detect different tea grades.

TABLE IV

PERFORMANCE OF THE AI-BASED TEA SORTING MODEL

| Parameter         | Value                                                 |
|-------------------|-------------------------------------------------------|
| Model             | ConvNeXtV2-Tiny (convnextv2_tiny.fcmae_ft_in22k_in1k) |
| Dataset           | Processed Tea Grade Image Dataset                     |
| Tea Grade Classes | BM, BOP, BP, BROKEN_TEA, DUST, FANNING_2, PF, PW_DUST |
| Average Precision | 99.7%                                                 |
| Average Recall    | 99.7%                                                 |
| Average F1-Score  | 99.7%                                                 |

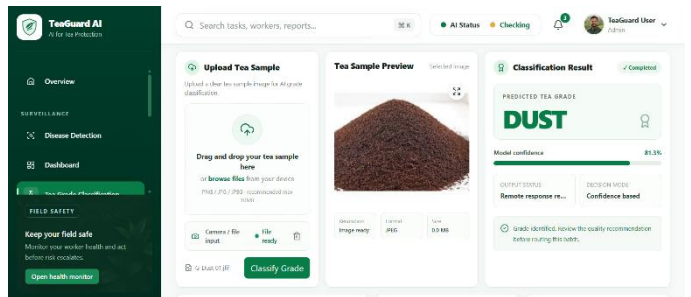


Fig. 6 AI-Based Tea Sorting and Grading

The results show that the proposed system works well when compared with the studies that were talked about in the literature review. Earlier research mostly looked at tasks like managing workers finding diseases predicting the weather or sorting tea leaves.. This study brings all four of those tasks together in one smart system and still gets good results. These results show how useful the system is, for making decisions helping workers improving tea quality and increasing how much tea is produced in Sri Lanka.

## V. CONCLUSION

This research presented an Intelligent Decision Support System for Sri Lankan Tea Estate Management to improve tea estate operations through a single web based platform. The proposed system integrates labor management, tea leaf disease detection, tea specific weather intelligence, and AI based tea sorting and grading. The experimental results showed that the developed models achieved good performance and can support accurate and timely decision making. The system helps improve worker welfare, detect tea leaf diseases at an early stage, provide weather based recommendations, and automate tea grading. Future work will focus on expanding the datasets, improving model performance, integrating additional IoT devices, and developing a mobile version to make the system more practical for tea estate management.

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